

Program: MapReduce Word Frequency Count (Python using Hadoop Streaming)

mapper.py

```
#!/usr/bin/env python3
import sys

# Read input line by line
for line in sys.stdin:
    line = line.strip()
    words = line.split()
    for word in words:
        print(f'{word}\t1')
```

reducer.py

```
#!/usr/bin/env python3
import sys

current_word = None
current_count = 0

# Input comes from standard input
for line in sys.stdin:
    word, count = line.strip().split('\t')

    try:
        count = int(count)
    except ValueError:
        continue

    if current_word == word:
        current_count += count
    else:
        if current_word:
            print(f'{current_word}\t{current_count}')
        current_word = word
        current_count = count

# Output the last word
if current_word == word:
    print(f'{current_word}\t{current_count}')
```

Input File (input.txt):

```
hello world welcome to the world of big data hello
```

Execution Commands:

```
# Put input file into HDFS
hadoop fs -put input.txt /input

# Run MapReduce job
hadoop jar /usr/lib/hadoop-mapreduce/hadoop-streaming.jar \
-input /input/input.txt \
-output /output \
-mapper mapper.py \
-reducer reducer.py

# View output
hadoop fs -cat /output/part-00000
```

Expected Output:

big	1
data	1
hello	2
of	1
the	1
to	1
welcome	1
world	2